

1.1 Analytical Question: The Duality of PCA Objectives

Problem statement (unchanged).

Provide a formal proof demonstrating that for a centered dataset, the objective of maximizing the variance of the projected data is equivalent to the objective of minimizing the mean squared reconstruction error. Supplement your proof with a geometric argument, possibly using a diagram, that illustrates this relationship in a simple 2D or 3D case. Explain why this duality is significant for interpreting PCA's function.

Solution (reformatted):

The duality of PCA's objectives is a cornerstone of its formulation, revealing that the seemingly different goals of maximizing information preservation (variance) and minimizing representational error (reconstruction loss) are, in fact, two sides of the same coin.

Formal proof

Let the dataset be represented by a set of N data points $\{\mathbf{x}_n\}_{n=1}^N$, where each $\mathbf{x}_n \in \mathbb{R}^D$. Assume the data is centered, i.e.

$$\frac{1}{N} \sum_{n=1}^N \mathbf{x}_n = \mathbf{0}.$$

Project the data onto an M -dimensional subspace ($M < D$) spanned by an orthonormal basis $\{\mathbf{u}_i\}_{i=1}^M$. For simplicity consider $M = 1$, i.e. projection onto the line spanned by the unit vector $\mathbf{u}_1$.

(a) Maximizing projected variance

The projection of $\mathbf{x}_n$ onto $\mathbf{u}_1$ is the scalar $\mathbf{u}_1^\top \mathbf{x}_n$. Because the data is centered, the mean of these projections is zero:

$$\frac{1}{N} \sum_{n=1}^N \mathbf{u}_1^\top \mathbf{x}_n = \mathbf{u}_1^\top \left(\frac{1}{N} \sum_{n=1}^N \mathbf{x}_n \right) = 0.$$

Hence the variance of the projected data is

$$\text{Var}_{\text{proj}} = \frac{1}{N} \sum_{n=1}^N (\mathbf{u}_1^\top \mathbf{x}_n)^2 = \mathbf{u}_1^\top \left(\frac{1}{N} \sum_{n=1}^N \mathbf{x}_n \mathbf{x}_n^\top \right) \mathbf{u}_1 = \mathbf{u}_1^\top \mathbf{S} \mathbf{u}_1,$$

where $\mathbf{S} = \frac{1}{N} \sum_{n=1}^N \mathbf{x}_n \mathbf{x}_n^\top$ is the (population-style) sample covariance matrix. The objective is to maximize $\mathbf{u}_1^\top \mathbf{S} \mathbf{u}_1$ subject to $\mathbf{u}_1^\top \mathbf{u}_1 = 1$. This is the Rayleigh quotient maximization: the maximizer is the eigenvector of $\mathbf{S}$ corresponding to its largest eigenvalue.

(If using the unbiased estimator $\mathbf{S} = \frac{1}{N-1} \sum \mathbf{x}_n \mathbf{x}_n^\top$ the same eigenvectors result; only scale changes.)

(b) Minimizing mean squared reconstruction error (MSRE)

The projection (reconstruction) of $\mathbf{x}_n$ onto the line spanned by $\mathbf{u}_1$ is

$$\tilde{\mathbf{x}}_n = (\mathbf{u}_1^\top \mathbf{x}_n) \mathbf{u}_1.$$

The reconstruction error for the point is $\mathbf{e}_n = \mathbf{x}_n - \tilde{\mathbf{x}}_n$. The mean squared reconstruction error is

$$J = \frac{1}{N} \sum_{n=1}^N \|\mathbf{x}_n - \tilde{\mathbf{x}}_n\|^2.$$

Expand the squared norm:

$$\begin{aligned} \|\mathbf{x}_n - \tilde{\mathbf{x}}_n\|^2 &= \mathbf{x}_n^\top \mathbf{x}_n - 2(\mathbf{u}_1^\top \mathbf{x}_n)(\mathbf{x}_n^\top \mathbf{u}_1) + (\mathbf{u}_1^\top \mathbf{x}_n)^2 (\mathbf{u}_1^\top \mathbf{u}_1) \\ &= \mathbf{x}_n^\top \mathbf{x}_n - (\mathbf{u}_1^\top \mathbf{x}_n)^2, \end{aligned}$$

since $\mathbf{u}_1^\top \mathbf{u}_1 = 1$. Averaging over n ,

$$J = \frac{1}{N} \sum_{n=1}^N \mathbf{x}_n^\top \mathbf{x}_n - \frac{1}{N} \sum_{n=1}^N (\mathbf{u}_1^\top \mathbf{x}_n)^2.$$

The first term,

$$\frac{1}{N} \sum_{n=1}^N \mathbf{x}_n^\top \mathbf{x}_n,$$

is the total variance of the data (a constant with respect to $\mathbf{u}_1$). The second term equals Var_{proj} . Therefore

$$J = \text{Total Variance} - \text{Var}_{\text{proj}}.$$

Minimizing J is equivalent to maximizing Var_{proj} . This proves the equivalence of the two objectives for $M = 1$; the argument extends to $M > 1$ by projecting onto an M -dimensional orthonormal subspace and summing over components.

Geometric argument

Take a single centered point $\mathbf{x} \in \mathbb{R}^2$ and a unit direction $\mathbf{u}_1$. Decompose $\mathbf{x}$ into two orthogonal parts:

$$\mathbf{x} = \tilde{\mathbf{x}} + \mathbf{e}, \quad \text{where} \quad \tilde{\mathbf{x}} = (\mathbf{u}_1^\top \mathbf{x}) \mathbf{u}_1, \quad \mathbf{e} = \mathbf{x} - \tilde{\mathbf{x}}.$$

By the Pythagorean theorem,

$$\|\mathbf{x}\|^2 = \|\tilde{\mathbf{x}}\|^2 + \|\mathbf{e}\|^2.$$

Summing over all data points and dividing by N ,

$$\frac{1}{N} \sum_{n=1}^N \|\mathbf{x}_n\|^2 = \frac{1}{N} \sum_{n=1}^N \|\tilde{\mathbf{x}}_n\|^2 + \frac{1}{N} \sum_{n=1}^N \|\mathbf{e}_n\|^2,$$

i.e.

$$\text{Total Variance} = \text{Projected Variance} + \text{Reconstruction Error}.$$

Since the total variance is fixed for the dataset, increasing the projected variance necessarily decreases the reconstruction error by the same amount — hence the duality.

Significance of the duality

- **Information-theoretic view (maximize variance):** PCA finds directions that preserve the most variability (assumed to carry most of the signal).
- **Geometric/compression view (minimize error):** PCA finds the best low-dimensional approximation in least-squares sense.

That these two views yield the same solution (the leading eigenvectors of the covariance) gives PCA both a statistically meaningful and a geometrically optimal interpretation.

1.2 Numerical Problem: Manual PCA Calculation

Given dataset

$$\mathbf{X} = \begin{pmatrix} 1 & 3 \\ 3 & 5 \\ 5 & 1 \end{pmatrix}.$$

Rows are data points: $\mathbf{x}_1 = (1, 3)^\top$, $\mathbf{x}_2 = (3, 5)^\top$, $\mathbf{x}_3 = (5, 1)^\top$.

Step 1 — Mean and centering

Feature means:

$$\mu_1 = \frac{1 + 3 + 5}{3} = 3, \quad \mu_2 = \frac{3 + 5 + 1}{3} = 3.$$

Mean vector $\boldsymbol{\mu} = (3, 3)^\top$.

Centered data (subtract mean from each row):

$$\mathbf{X}_{\text{centered}} = \begin{pmatrix} 1-3 & 3-3 \\ 3-3 & 5-3 \\ 5-3 & 1-3 \end{pmatrix} = \begin{pmatrix} -2 & 0 \\ 0 & 2 \\ 2 & -2 \end{pmatrix}.$$

Step 2 — Covariance matrix

Using the unbiased estimator $\mathbf{S} = \frac{1}{N-1} \mathbf{X}_{\text{centered}}^T \mathbf{X}_{\text{centered}}$ with $N = 3$:

First compute $\mathbf{X}_{\text{centered}}^T \mathbf{X}_{\text{centered}}$:

$$\begin{pmatrix} -2 & 0 & 2 \\ 0 & 2 & -2 \end{pmatrix} \begin{pmatrix} -2 & 0 \\ 0 & 2 \\ 2 & -2 \end{pmatrix} = \begin{pmatrix} (-2)^2 + 0^2 + 2^2 & (-2)(0) + 0(2) + 2(-2) \\ 0(-2) + 2(0) + (-2)2 & 0^2 + 2^2 + (-2)^2 \end{pmatrix} = \begin{pmatrix} 8 & -4 \\ -4 & 8 \end{pmatrix}.$$

Then

$$\mathbf{S} = \frac{1}{3-1} \begin{pmatrix} 8 & -4 \\ -4 & 8 \end{pmatrix} = \begin{pmatrix} 4 & -2 \\ -2 & 4 \end{pmatrix}.$$

Step 3 — Eigenvalues and first principal component

Find eigenvalues of $\mathbf{S}$:

$$\det(\mathbf{S} - \lambda \mathbf{I}) = \det \begin{pmatrix} 4 - \lambda & -2 \\ -2 & 4 - \lambda \end{pmatrix} = (4 - \lambda)^2 - 4 = 0.$$

So

$$(4 - \lambda)^2 = 4 \quad \Rightarrow \quad 4 - \lambda = \pm 2.$$

Thus eigenvalues are

$$\lambda_1 = 4 + 2 = 6, \quad \lambda_2 = 4 - 2 = 2.$$

The largest eigenvalue is $\lambda_1 = 6$.

Find eigenvector for $\lambda_1 = 6$:

$$(\mathbf{S} - 6\mathbf{I})\mathbf{v} = \begin{pmatrix} 4 - 6 & -2 \\ -2 & 4 - 6 \end{pmatrix} \mathbf{v} = \begin{pmatrix} -2 & -2 \\ -2 & -2 \end{pmatrix} \mathbf{v} = \mathbf{0}.$$

This gives $-2v_1 - 2v_2 = 0$ so $v_1 = -v_2$. A (unnormalized) eigenvector is $(1, -1)^T$. Normalizing:

$$\mathbf{u}_1 = \frac{1}{\sqrt{1^2 + (-1)^2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Step 4 — Project original (centered) data onto the first PC

Compute $\mathbf{Z} = \mathbf{X}_{\text{centered}} \mathbf{u}_1$. With $\mathbf{u}_1 = \frac{1}{\sqrt{2}}(1, -1)^T$,

$$\mathbf{Z} = \begin{pmatrix} -2 & 0 \\ 0 & 2 \\ 2 & -2 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} -2(1) + 0(-1) \\ 0(1) + 2(-1) \\ 2(1) + (-2)(-1) \end{pmatrix} = \begin{pmatrix} -2/\sqrt{2} \\ -2/\sqrt{2} \\ 4/\sqrt{2} \end{pmatrix} = \begin{pmatrix} -\sqrt{2} \\ -\sqrt{2} \\ 2\sqrt{2} \end{pmatrix}.$$

So the 1D representations (projections) of the three data points on the first principal component are approximately

$$[-1.414, -1.414, 2.828].$$